ANALYTICAL CUMULATIVE EXAM
December 7, 1996

Professor Kim Prather from the University of California, Riverside, gave a talk in our department on December 2, 1996. This cume is based upon that talk.

1. (10 pts.) What does ATOFMS stand for?

2. (10 pts.) What kinds of samples does Professor Prather analyze? How does she categorize them?

3. (40 pts.) What kind of instrumentation does Professor Prather use to analyze the samples? Describe the instrumentation in detail. What different kinds of properties of substances does it measure? How does it measure them?

4. (30 pts.) Describe another analytical method (such as the one used by Prather's collaborators) for analyzing such samples. How would it work? How would it compare (in sensitivity, accuracy, precision, cost, time, etc.) to Prather's analytical tool?

5. (10 pts.) What were some of the results found when analyzing samples at the time of the July 4th Holiday?